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RESEARCH

Can knowledge of the spatial field trend improve phenotypic prediction accuracy through informed choice of experimental design?

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Abstract Environmental variation is an important factor in crop yield variability, especially in large agricultural experiments such as breeding trials. This study explores whether prior knowledge of the spatial field trends can enhance phenotypic prediction accuracy by choice of experimental design and statistical analysis. Using data from three breeding trials and a simulation study with eight field trends, we compared five experimental designs and six statistical models. The results showed that combining field trend knowledge with an appropriate analysis strategy can improve predictive performance. There was no universally optimal design; but among the designs considered, the row-column design performed best for most field trends with a high degree of spatial variation. Statistical models correcting for spatial variation consistently reduced bias and increased correlation between predicted and true yield. This work emphasizes the need to integrate spatial information in the decision of both the design and analysis of breeding trials to boost prediction accuracy.

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Introduction

The greatest cause of variability in crop yield is environmental variation (Mackay et al. 2019). Still, comparative experiments in agriculture are often located on fields with some degree of variation, which especially becomes an issue for large experiments such as breeding trials. Even though there is a clear aim to find homogeneous fields, breeders face the challenge of potentially high background variation due to the high number of genotypes to be compared.

Good trait estimation and accordingly genotype ranking rely on effective control of the field variation. This is commonly addressed by choice of experimental design, statistical analysis, or a combination of both (Smith et al. 2005; van Eeuwijk et al. 2019). Without proper treatment, spatial effects and heterogeneity can reduce the quality of the genotype ranking and phenotypic predictions (Mao et al. 2020).

When genotypes are regarded as random, as often done in the statistical analysis of breeding trials, it has been shown that the optimal experimental design is not independent of the genetic correlation structure (Bueno Filho and Gilmour 2003, 2007). Optimal designs, however, often depend on the covariance to be known beforehand. A scenario that in practice collides with the lacking knowledge of the genetic variance before running a trial combined with the breeders' preference for robust and easy-to-use procedures. Further, a design optimized for one trait may not necessarily be optimal for another trait. Accordingly, it is desirable to choose general-purpose designs that are likely to perform well across a wide range of conditions (Piepho and Williams 2006).

Depending on the type and degree of spatial trend, blocked experimental designs may provide efficient trait estimates by capturing part of the spatial variation (Edmondson 2005). The most commonly used design in agricultural experiments is the randomized complete block design (RCBD) that is simple, flexible, and robust (Casler 2015). The RCBD can be effective against a range of nuisance effects such as residual effects of previous treatments, differences in sowing or harvest dates, management zones, or locations (Bailey 2008). However, the sources of variability within replicate blocks are often unknown and may appear on different scales, and for breeding experiments with many genotypes, the RCBD may fail to provide proficient control across the entire experimental field, due to the large block sizes needed for such trials. Also, there is no reason to believe that a single block size is sufficient to account for the variability across a range of different scales in large replicate blocks (Edmondson 2020). It is therefore common practice to divide large replicate blocks into smaller incomplete blocks for improved within-block homogeneity (Bailey 2008). Two such incomplete block designs are the row-column (John and Eccleston 1986) and the alpha designs (Patterson and Williams 1976). Especially the alpha designs, which are partially balanced resolvable incomplete block designs, are often suggested for and widely employed in breeding trials because they allow for the incorporation of a large number of genotypes (Piepho and Williams 2006; Oehlert 2010). Row-column designs are advantageous because they utilize blocking in both directions, thereby reducing the risk of blocking in the wrong dimension, which is a potential issue with one-dimensional designs when there is no prior knowledge of the field trend (Zystro et al., 2019). Moreover, for traits of high heritability, like yield, it has been found that the greatest response to selection can come from experiments in which most entries occur only once, while a few control entries are replicated many times, i.e., the augmented design (Mackay et al. 2019).

To take full advantage of block designs, whether complete or incomplete, block sizes and placement should be guided by knowledge of the local field variability. Even though field conditions and block effects will vary from year to year due to a range of factors including management and weather, understanding the general field variation may enhance the decision-making process for selecting an appropriate experimental design (Wilkinson et al. 1983). Richter and Kroschewski (2012) demonstrated similar field trends in five consecutive years based on uniformity trials with five different crops, supporting the existence of general field trends despite variations.

Statistical analysis should always go hand in hand with the experimental design (Jensen et al. 2018) such that any kind of blocking used in the experimental design is accounted for in the data analysis, for example using a linear mixed model (LMM) (McCulloch et al. 2008), but the classic LMMs are not good at capturing the within block variation. However, spatial control of additional variation not captured by the experimental design and randomization can be achieved at the analysis step (Rodríguez-Álvarez et al. 2018; Borges et al. 2019). Therefore, different statistical models that include some kind of additional adjustment for spatial correlations have been proposed, for example using variance models.

Incorporating both random blocks and spatial smoothers within a single model is likely to capture the underlying smooth patterns within and across blocks, all while accommodating variations from smoothness within each block. Bernal-Vasquez et al. (2014) found, that fitting a row-column model, i.e. a model that includes an effect for each row and each column in the experimental layout, can account for a substantial proportion of the phenotypic heterogeneity from spatial effects. Edmondson 2020 found that smoothing alone was not sufficient to account for trends between columns in a crossed row-column design, but that a combination of smoothing and random blocks was required to account for trends within rows and columns. This strategy can in principle be employed in any field block design, emphasizing the importance of a well-structured (block) design during the planning phase to optimize the flexibility of model fitting during the analysis phase (Edmondson 2020). Rodriguez-Álvarez et al. (2018) used a bivariate smooth surface based on P-splines combined with random blocks to model the spatial effects capturing both large- and small-scale dependencies and Mao et al. (2020) used a Gaussian random field model, with an additive covariance structure incorporating genotype, subpopulation and spatial effects and illustrated an improvement in genomic selection by adjusting for the spatial effects.

While the sophistication of statistical models enhances the ability to adjust for spatial effects, their evaluation has predominantly relied on field trends simulated using one of the models under consideration, implying the inclusion of the true model. Alternatively, and even less realistically, these evaluations often disregard field trends beyond block effects. However, it remains unexplored whether these models retain superior predictive performance across a variety of field trends and when combined with different common experimental designs, and more so, whether design and analysis strategies can be improved based on prior knowledge of the underlying field trend.

Inspired by Cooper et al. (2014), who suggested that spatial patterns obtained using precision agriculture technology can be utilized to improve the design of experiments and determine suitable spatial corrections for each experiment, the aim of this study is to examine if preceding knowledge of the field trend can provide the basis for an informed decision on design and analysis strategy for breeding trials. Based on data from three breeding trials and an extensive simulation study with eight different field trends, we investigate the ability of different experimental designs and analysis strategies to appropriately adjust for the field variation. Furthermore, we compare the performance of established statistical models with and without adjustment for spatial variation across designs and field trends. We hypothesize that one experimental design does not fit all field trends, but that much can be gained by an appropriate choice of analysis strategy.

Material and methods

Data example

Data came from three breeding trials with winter barley lines from Sejet Plant Breeding. The field trials were carried out in 2020, 2021, and 2022 and were located in the Jutland region close to the breeding station near Horsens, Denmark. All three trials consisted of multiple sub-experiments laid out as alpha lattice designs. Each sub-experiment had three repetitions, each consisting of six incomplete (mini) blocks, resulting in a total of 800, 892, and 1167 lines including check lines, in 2020, 2021 and 2022, respectively.

During each growth season, the experiments were imaged using an unmanned aerial vehicle (UAV). A DJI Phantom 3 drone (DJI, Shenzhen, China, www.dji.com) was used in season 2020, whereas DJI Mavic 2 professional (DJI, Shenzhen, China, www.dji.com) was used in season 2021 and 2022. The flight altitude was 60 m and the white balance in the camera was fixed before flying. The ground sampling distance was 2.6 cm/pixel for the DJI Phantom 3 and 1.4 cm/pixel for the Mavic 2 professional. The orthomosaic generation and data extraction were carried out using DroneDeploy version 4.113. 0 (DroneDeploy, San Francisco, USA) and the normalized excess green index (NExG) was extracted for each experimental plot. At maturity, each plot was harvested for grain yield.

Field trends

Eight different field trends were generated for a simulation study (Fig. 1). The field trends considered were limited to represent spatial variation due to environmental factors such as variation in soil fertility and water availability, but not the controllable management factors.

Field trends 1–4 were generated as a Gaussian mixture of 2-dimensional normal distributions. These were made to illustrate general small- and large-scale trends across an experimental field, e.g. fields containing sand pockets or areas with insufficient draining. Field trend 5 was designed to illustrate the effect of compacted soil from tractor tracks. Trend 6 was the perfect but often unrealistic homogeneous field and trends 7 and 8 were based on the field trends from two of the breeding trials from the data example. The effects of all field trends were centered and scaled to have zero mean and unit variance.

Experimental designs

Five different designs were considered: a randomized complete block design (RCBD), an alpha lattice

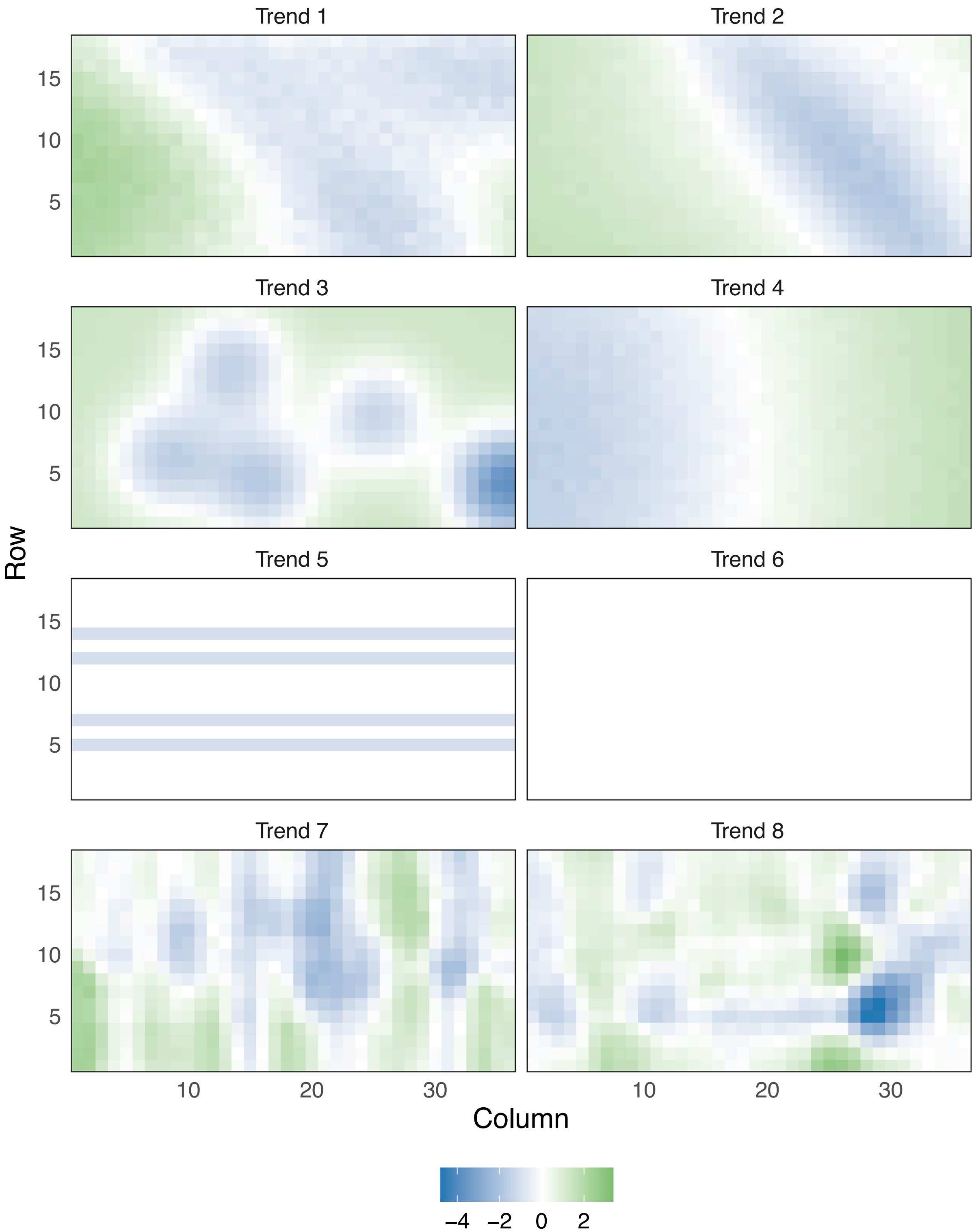


Fig. 1 The eight field trends used in the simulation study. Field trends 1–6 were simulated data while trends 7–8 were based on the breeding trials in the data example

design, a row-column design, an augmented RCBD design, and an augmented row-column design. Each design was applied to a field with a fixed layout consisting of a grid of 18 rows and 36 columns, i.e., a total of 648 plots. To fit this layout, each design was applied as (sub-) experiments with the field, where the experiments were independent in the sense that only check lines were allocated to plots across experiments. The remaining lines only appeared in one experiment. The designs were applied in different configurations of number of experiments, number of repetitions, and number of incomplete (mini) blocks per repetition for the alpha design. In addition, different numbers of check lines were considered with corresponding room for fewer new lines to be tested (Table 1).

For placement of experiments with each layout in the field and examples of design randomizations, see Figs. S1-S43 in the supplementary material.

Statistical models for field trials

When analyzing data from a field trial, the analysis should account for experimental design and management as well as the underlying spatial effects. There are several statistical approaches that may be used for this purpose. Here, six different approaches with a varying degree of potential for capturing field trends were considered.

Note that this work only considered the case where selection was based on the entire set of entries in the data set, as opposed to selection within groups of lines.

Linear mixed models

Two different linear mixed models were considered, a linear mixed model with nested design structure:

Table 1 Experimental design configurations considered in the simulation study

Design	Exp	Exp layout (RxC)	Rep within Exp	Blocks per Rep (Size)	Checks	Lines per Exp	Lines in total
RCBD	18	(6×6)	2	1 (18)	3, 4 or 5	18	273, 256, 239
RCBD	18	(6×6)	3	1 (12)	3, 4 or 5	12	165, 148, 131
RCBD	12	(6*9)	3	1 (18)	3, 4 or 5	18	183, 172, 161
RCBD	8	(9*9)	3	1 (27)	3, 4 or 5	27	195, 188, 181
Alpha	18	(6×6)	2	6 (3)	3, 4 or 5	18	273, 256, 239
Alpha	18	(6×6)	3	4 (3)	3, 4 or 5	12	165, 148, 131
Alpha	12	(6*9)	3	6 (3)	3, 4 or 5	18	183, 172, 161
Alpha	8	(9*9)	3	9 (3)	3, 4 or 5	27	195, 188, 181
Row-col	18	(6×6)	2	6R*3C (3,6)	3, 4 or 5	18	273, 256, 239
Row-col	18	(6×6)	3	6R*2C (2,6)	3, 4 or 5	12	165, 148, 131
Row-col	12	(6*9)	3	6R*3C (3,6)	3, 4 or 5	18	183, 172, 161
Row-col	8	(9*9)	3	9R*3C (3,9)	3, 4 or 5	27	195, 188, 181
Aug. block	18	(6×6)	2 (checks)	1 (18)	3, 4 or 5	33, 32, 31	543, 508, 473
Aug. block	18	(6×6)	3 (checks)	1 (12)	3, 4 or 5	30, 28, 26	489, 436, 383
Aug. block	12	(6*9)	3 (checks)	1 (18)	3, 4 or 5	48, 46, 44	543, 508, 473
Aug. block	8	(9*9)	3 (checks)	1 (27)	3, 4 or 5	75, 73, 71	579, 556, 533
Aug. row-col	18	(6×6)	2 (checks)	6R*3C (3,6)	3, 4 or 5	33, 32, 31	543, 508, 473
Aug. row-col	18	(6×6)	3 (checks)	6R*2C (2,6)	3, 4 or 5	30, 28, 26	489, 436, 383
Aug. row-col	12	(6*9)	3 (checks)	6R*3C (3,6)	3, 4 or 5	48, 46, 44	543, 508, 473
Aug. row-col	8	(9*9)	3 (checks)	9R*3C (3,9)	3, 4 or 5	75, 73, 71	579, 556, 533

The field layout was divided into a given number of experiments (Exp), each containing 2 or 3 repetitions (Rep), with a fixed number of (mini) blocks (Blocks) of a certain size within each repetition. Varying number of check lines (Checks) were used across all experiments in the field resulting in different number of lines in each experiment and total number of lines tested across the field including the check lines

$$Y = X\beta + Z_{\text{Exp}}c_{\text{Exp}} + Z_{\text{Exp:Rep}}c_{\text{Exp:Rep}} + Z_{\text{Exp:Rep:Block}}c_{\text{Exp:Rep:Block}} + \epsilon \quad (1)$$

and a linear mixed model with rows and columns as random effects:

$$Y = X\beta + Z_{\text{Row}}c_{\text{Row}} + Z_{\text{Col}}c_{\text{Col}} + \epsilon, \quad (2)$$

where Y was the vector of responses (yield); β was the coefficients for the fixed effects, $c_{\text{Exp}}, c_{\text{Exp:Rep}}, c_{\text{Exp:Rep:Block}}, c_{\text{Row}}, c_{\text{Col}}$ were random effect coefficients for experiment, repetition nested in experiment, block nested in repetition nested in experiment, row and column, respectively. $X, Z_{\text{Exp}}, Z_{\text{Exp:Rep}}, Z_{\text{Exp:Rep:Block}}, Z_{\text{Row}}, Z_{\text{Col}}$ were the corresponding design matrices for the fixed and random effects. All random effects, c_{x}, were assumed to follow a normal distribution, $c_{\text{x}} \sim N(0, \sigma_x^2)$ and to be independent of the residuals,$\epsilon \sim N(0, \sigma^2)$. For the nested random effects approach, only random effects corresponding to the associated design were considered, e.g., for an alpha lattice design, experiment, replicate and block were included as random effects, while analyses from an RCBD would only include experiment and replicate as random effects. The linear mixed models, (1) and (2), would be expected to capture part of the spatial trend in the field, but to a degree that would depend on the trend, the strength of the trend and location of blocks relative to the field trend.

Linear mixed model with a correlation structure

The next approaches were adding to the linear mixed model approach by allowing for correlated noise. Two different correlation structures were considered, a linear and a Gaussian correlation structure. Both correlation structure models were built on top of the linear mixed model with nested random effects (1), by allowing a linear correlation structure in both rows and columns, such that.

$$\text{cor}\left(x_{\text{row,col}}, x_{\text{row}+i,\text{col}}\right) = \text{cor}\left(x_{\text{row,col}}, x_{\text{row,col+i}}\right) = 1 - \left(\frac{i}{d}\right) \text{ for } i < d \quad (3)$$

or a Gaussion correlation structure, such that

$$\text{cor}\left(x_{\text{row,col}}, x_{\text{row}+i,\text{col}}\right) = \text{cor}\left(x_{\text{row,col}}, x_{\text{row,col+i}}\right) = \exp\left(-\left(\frac{i}{d}\right)^2\right) \quad (4)$$

for a range, d. In the simulations below, d was set to 3.

Linear mixed model with P-splines modelling of the spatial variation in two dimensions

The final two approaches considered were based on a linear mixed model combined with a two-dimensional smooth surface estimated using P-splines to model the spatial variation (Rodríguez-Álvarez et al. 2018):

$$Y = f(u, v) + X\beta + Z_{\text{Exp}}c_{\text{Exp}} + Z_{\text{Exp:Rep}}c_{\text{Exp:Rep}} + Z_{\text{Exp:Rep:Block}}c_{\text{Exp:Rep:Block}} + \epsilon \quad (5)$$

Or

$$Y = f(u, v) + X\beta + Z_{\text{Row}}c_{\text{Row}} + Z_{\text{Col}}c_{\text{Col}} + \epsilon, \quad (6)$$

for the two versions with either nested random effects (5) or rows and columns as random effects (6). $f(u, v)$ was a bivariate surface modeled as a tensor product P-spline mixed effects decomposition incorporating smooth trends along the rows,u, and columns, v, linear-by-smooth interaction trends, as well as smooth-by-smooth interaction trends (Rodríguez-Álvarez et al. 2018).

Statistical analysis

Data example

For each of the three breeding trials, the underlying field trend was estimated using drone and yield data separately. A linear mixed model was fitted to the yield data and to UAV-based NExG data for each time point, with lines as fixed effect and experiment, repetition and blocks as random effects. It was assumed that any systematic trend in the residuals not captured by repetition and blocks could be ascribed to the underlying field trend. Accordingly, a K-nearest neighbor (KNN) smoothing with K = 100 of the residuals was used to obtain the field trend.

The estimated field trends in NExG obtained using UAV during the growth season were compared to the field trend observed in the yield using Spearman correlations. Three types of correlations between NExG values and yield were considered based on plot values; (1) Correlations based on raw data, i.e. observed NExG correlated to observed yield; (2) Correlations based on residuals from a linear mixed model adjusting for experiments, repetitions, and blocks; (3) Field trends based on KNN on the residuals with K = 100.

Additionally, data from each of the three breeding trials were analysed using the six different statistical models outlined above to estimate the mean yield for each line. Results from the different analysis strategies were compared in terms of Spearman correlations to judge whether different analysis strategies would result in a different ranking of breeding lines.

Simulations

A simulation study was conducted to compare the performance of the different design and analysis strategies for the different field trends. For each scenario, the line-specific values were sampled from a normal distribution with mean 100 and standard deviation 10 to provide variability among genotypes. The lines were then randomly distributed to the field layout based on experimental design, giving a sample size of 648 for each simulated data set. The experimental design was independently randomized for each experiment in the field layout. To have variation within genotypes, Gaussian noise was added, by sampling from a normal distribution with mean 0 and standard deviation 1.5. As a final step, the field trend was included by assuming an additive effect, corresponding to the underlying assumption of most statistical models. The additive effect was obtained by multiplying the field trend mask with 5 before adding the effect to the field layout. In addition, two sensitivity analyses were made. In the first, the effect of a lower heritability through an increased spatial variation was investigated. The higher spatial variation was achieved by instead multiplying the field trend mask with 10 before adding it to the field layout. In the second sensitivity analysis representing a scenario not following the assumptions of the underlying statistical models considered, a multiplicative field trend was considered. This was made by modifying the field trend masks to be centered around 1 after which it was multiplied to the field layout.

For each combination of field trend and experimental design configuration (Table 1), 1000 data sets were simulated.

Each data set was analysed using six different statistical models; a linear mixed model with nested random effects according to the experimental design; a linear mixed model with rows and columns as random effects; a linear mixed model with nested random effects and a linear correlation structure; a linear mixed model with nested random effects and a Gaussian correlation structure; a linear mixed model with P-splines modelling of the spatial variation (SpATS) model with nested random effects; and a SpATS model with rows and columns included as random effects.

Results were summarized as absolute relative bias in estimated yield and in terms of the Spearman rank correlation between the true and the estimated yield.

All analyses and simulations were carried out in R version 4.3.2 (R Core Team 2023). In particular the following extension packages were used: agricolae (de Mendiburu 2021) for randomizing RCBD, alpha and augmented designs, FielDHub for randomizing row-column and augmented row-column design (Murillo and Gezan 2024), lme4 (Bates et al. 2015) for fitting linear mixed models with nested and row-column random effects, nlme (Pinheiro and Bates 2000) for fitting linear mixed models with linear and Gaussian correlation structures and SpATS (Rodríguez-Álvarez et al. 2018) for fitting mixed models with P-splines.

Results

Evaluation of field trends from UAV and yield data

The correlations between UAV-obtained NExG and residuals and the yield varied over the growth season and between years (Fig. 2) with the highest correlation obtained during heading in 2020 (Fig. 2A), during flowering and end of winter in 2021 (Fig. 2B), and during winter in 2022 (Fig. 2C). In 2021 and 2022, there was a higher correlation between the residuals observed from the NExG model and the residuals from the yield model compared to the correlation between the measured NExG and yield data indicating that residual NExG were able to capture part of the unexplained variation in yield observed

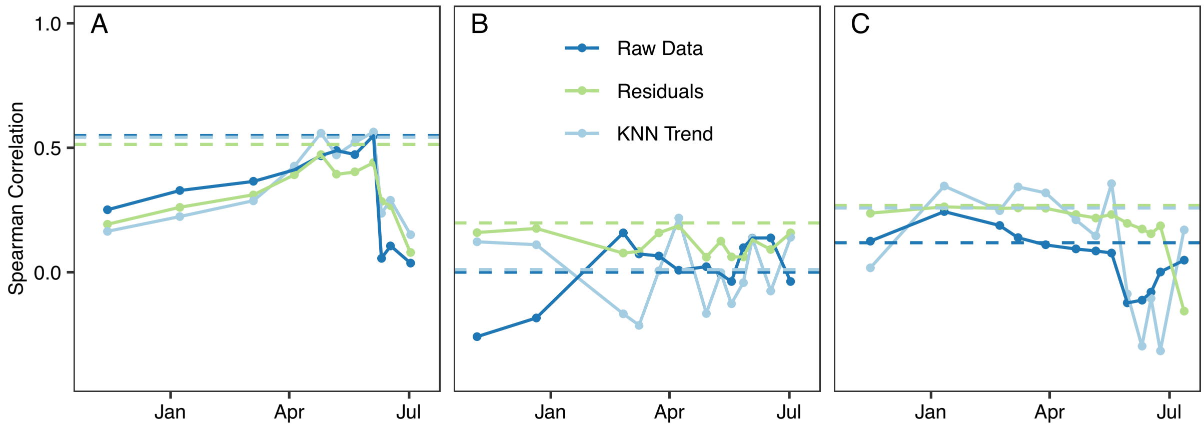


Fig. 2 Correlations between the field trend in yield and the field trend described by UAV-derived NExG at different time points during the growth season (full line) or the mean field trend described by UAV-derived NExG across the season (dashed line). Field trends were represented by raw data, residuals from a linear mixed model or residuals from a linear mixed model smoothed using K-nearest neighbors (KNN-trend). **A** Data from the 2020 trial. **B** Data from the 2021 trial. **C** Data from the 2022 trial

between replicates of the same line placed in different locations of a field. The underlying field trend, as expressed in yield, was best captured by the KNN-smoothed trend in 2022 (Fig. 2C) and to the lowest degree in 2021 (Fig. 2B). The mean residuals had a higher correlation to the yield residuals compared to the residuals from any individual time points in all three years indicating that taking the mean levelled out some of the variation. The increase in correlation by using mean values was also observed for the raw data as well as the KNN-smoothed trends in 2020 (Fig. 2).

Comparison of statistical models on breeding trials

Comparing the six different statistical models on the breeding trials revealed different results between analysis approaches. The estimated yields from the LMM with correlation structure and/or nested random effects were highly correlated in all three trials (Fig. 3). The results from the SpATS model with row-column adjustment were the ones standing out the most, especially for the 2020 trial.

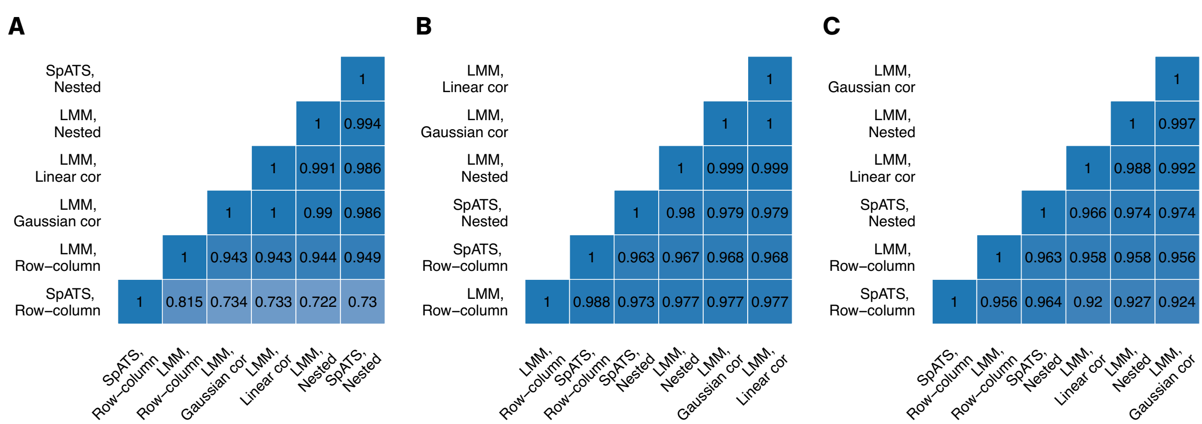


Fig. 3 Correlations between yield estimates for each line in the three field trials using six different analysis approaches. **A** Data from the 2020 trial. **B** Data from the 2021 trial. **C** Data from the 2022 trial

Simulation results: additive error structure

The lowest absolute relative bias was observed for field trend 6 followed by 4, while the overall highest bias was observed for field trend 8 followed by 7. For each of the designs, field trend 6 was consistently associated with the lowest bias and field trend 8 with the highest (Fig. 4).

The results on the correlations closely followed those on bias (Fig. 5). The highest correlation between true and estimated yield was observed for field trend 6, reaching a correlation of 0.99 on average across all scenarios. Among the more realistic field trends, a correlation of 0.98 was reached for field trend 4, while on average the lowest correlations were found for field trend 8 and 7 (0.90 and 0.92, respectively). Looking across experimental layout, number of check lines, and analysis approaches, the lowest correlation (0.83) was observed for field trend 8 with an augmented RCBD design, while the highest correlations ignoring field trend 6 were observed for field trend 4 and 5 with alpha and row-column designs (all > 0.98).

Among the designs considered, the row-column design led to the lowest bias and highest correlations for field trends 1, 2, 3, 4, 7 and 8, whereas the alpha design resulted in the lowest bias and highest correlations for field trends 5 and 6 (Figs. 4, 5).

Of the four field layouts considered, the layout with the smallest experiments (6 × 6) and three replicates resulted in the lowest bias and highest correlation while the layout with only two replicates resulted in the overall highest bias and lowest correlation, but with a very similar correlation for the field layout with the largest experiments (9 × 9). However, the field layout mainly had an effect for field trends 3, 7 and 8 and the augmented designs in terms of correlation.

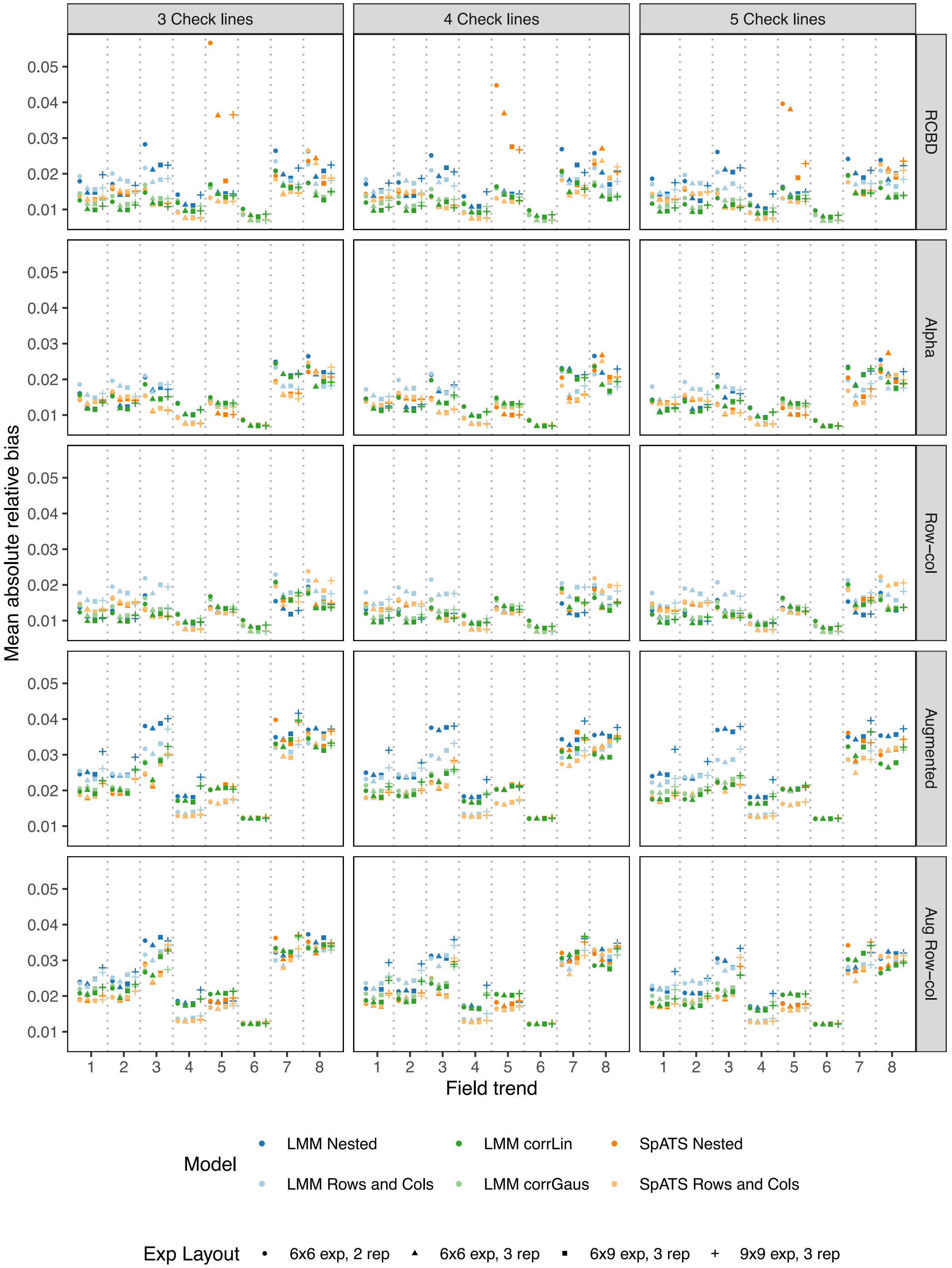
Overall, increasing the number of check lines from 3 to 5 only resulted in a 0.11% decrease in the absolute relative bias and an increase in correlation of 0.007. Increasing the number of check lines had the least effect for field trend 6, 5 and 4 in terms of both bias (all < 0.06%) and correlation (all < 0.002), while the highest effect was observed for field trend 3 in terms of bias (0.24%) and for field trend 8 in terms of correlation (0.017). The largest benefit from increasing the number of check lines was as expected obtained for both augmented designs.

The choice of analysis strategy had a smaller effect on the bias and correlation compared to the choice of experimental design. Models with spatial corrections besides what was accounted for by the randomization in the experimental design resulted in lower bias and higher correlations than models without any spatial correction. For field trends 1, 2 and 8, the LMM with linear correlation structure resulted in the lowest bias, for field trends 3 and 4, the two SPATS models performed best and very similar in terms of bias. The SpATS model with row-column adjustment performed best in terms of bias for field trend 7, while the LMM with row-column adjustment resulted in the lowest bias for field trend 5. All models performed similarly for field trend 6 with no spatial trend (Fig. 4). Among the considered modeling approaches, the LMM with nested design adjustment performed worst in terms of bias for most of the field trends.

Comparing the six different statistical models on the breeding trials revealed different results between analysis approaches. The estimated yields from the LMM with correlation structure and/or nested random effects were highly correlated in all three trials. The results from the SpATS model with row-column adjustment were the ones standing out the most, especially for the 2020 trial.

In terms of absolute relative bias, the optimal combination of design and analysis strategy for each field trend was for field trend 1, 2 and 8 the row-column design analysed using a LMM with a linear correlation structure (Fig. 4). Field trends 3, 4 and 7 also resulted in the lowest bias when using a row-column design, but with different analysis strategies (SpATS with nested design adjustment, SpATS with row-col adjustment, and LMM with nested design adjustment, respectively). The lowest bias for field trend 5 was found if an alpha design was combined with a SpATS model with nested design adjustment.

In terms of correlation, the SpATS model with row-column adjustment combined with the RCBD or the row-column design were among top 3 options for field trends 1, 2, 3, 5, and 7 (Fig. 5). For field trend 4, the best options were the alpha and row-column designs analysed using SpATS with nested design adjustment while the highest correlation for field trend 8 was found for the row-column design analysed using the LMM with nested design adjustment. No real differences in bias or correlation were


3 Check lines
4 Check lines
5 Check lines
RCBD
Alpha
Row–col
Augmented
Aug Row–col
Mean absolute relative bias
0.05
0.04
0.03
0.02
0.01
Field trend
1 2 3 4 5 6 7 8
Model
LMM Nested
LMM Rows and Cols
LMM corrLin
LMM corrGaus
SpATS Nested
SpATS Rows and Cols
Exp Layout
6x6 exp, 2 rep
6x6 exp, 3 rep
6x9 exp, 3 rep
9x9 exp, 3 rep

◀**Fig. 4** Mean absolute relative bias for different combinations of field trend, design and analysis strategy. Values are mean values over 1000 simulated data sets. Additive error structure

observed between designs and analysis strategies for field trend 6.

Sensitivity analyses with higher spatial variation

For each field trend, the results from the simulations with the higher spatial variation, i.e., lower heritability, were very similar to (highly correlated to) the results from the lower spatial variation presented above, but with the higher spatial variation resulting in higher bias and lower correlations (Figs. S44 and S45 in the supplementary material). The row-column design was still associated with the lowest bias and highest correlation. Doubling the variation was associated with a similar doubling of the effect of increasing the number of check lines from three to five (here a 0.22% decrease in bias). The best overall analysis strategy was now the LMM with linear correlation structure, however closely followed by the SpATS model with row-column adjustment as above.

Some outlying results were, however, observed for the SpATS model with nested design adjustment for field trend 5 when combined with the RCBD design. Where the general trend was that bias were doubled in size from low to high spatial variation (high to low heritability), these combinations resulted in an up to five times higher bias compared to the low spatial variation settings. However, the same settings were not associated with any additional decrease in correlation compared to the remaining combinations.

Sensitivity analyses with multiplicative error structure

With the multiplicative field trends, the absolute relative bias was again higher (Fig. S46). The higher bias was especially pronounced for field trend 5 where the mean bias across all scenarios reached 8.0% under the multiplicative field trend compared to 1.6% under the additive field trend. On the other hand, the lowest overall correlations were observed for field trend 8 (Fig. S47). Across field trends, the best performing design in terms of both bias and correlation was the row-column design followed by the alpha design.

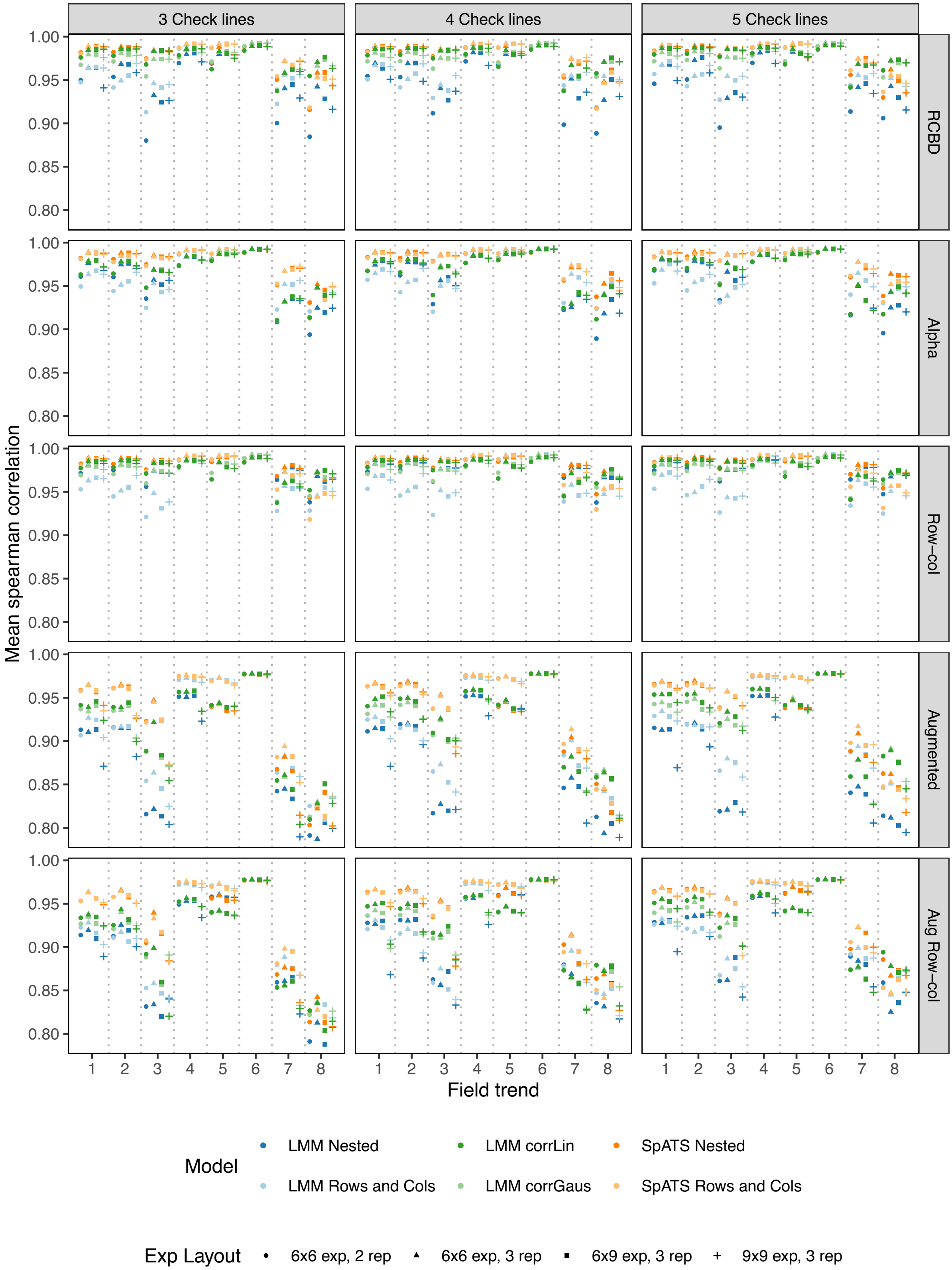
The effect of more check lines was larger for the multiplicative error structure than the absolute error structure, however increasing the number of check lines from 3 to 5 still resulted in a limited effect on relative bias (0.2%).

Across all field trends, the model strategy performing best was the LMM with linear correlation adjustment in terms of bias but still the SpATS model with row-column adjustment in terms of correlation. The potential improvement in relative bias by appropriate choice of design and analysis strategy was as big as 32.4% (field trend 5, app. 5% for trend 3 and 7) for the multiplicative error structure compared to approximately 2.5% (for field trends 3, 5, and 7) for the additive error structure.

Discussion

Value of realistic field trends in design and analysis evaluation

In most simulation studies examining the performance of statistical models adjusting for spatial effects, the underlying field trends are simulated based on one of the models examined in the associated study, i.e., the true model is among the ones considered (Selle et al. 2019; Mao et al. 2020). Alternatively, and even less realistic, field trends besides block effects are ignored (Piepho et al. 2006; Piepho and Williams 2006; Mackay et al. 2019). Inspired by the work of Hoefler et al. (2020), this study investigated field trends based on real data examples and on simulated trends resembling typical observed trends, but instead of only concluding on the performance of experimental designs and analysis approaches across trends, it was examined whether different types of underlying trends could lead to different recommendations. Despite a general trend that the SpATS model with row-column adjustment and a row-column design performed the best, it was also seen that some of the field trends resulted in lower bias when analysed using a different approach. This was the case for the field trends with very high local variation (Trends 1, 2 and 8), where the background variation was well-captured by LMMs with a linear correlation structure. However, in line with other studies, and for all field trends, it was found that estimates of the entries can be improved by accounting for spatial


3 Check lines
4 Check lines
5 Check lines
RCBD
Alpha
Row−col
Augmented
Aug Row−col
Mean spearman correlation
1.00
0.95
0.90
0.85
0.80
1 2 3 4 5 6 7 8
Field trend
Model
LMM Nested
LMM corrLin
SpATS Nested
LMM Rows and Cols
LMM corrGaus
SpATS Rows and Cols
Exp Layout
6x6 exp, 2 rep
6x6 exp, 3 rep
6x9 exp, 3 rep
9x9 exp, 3 rep

◂**Fig. 5** Spearman correlation to true yield for different combinations of field trend, design and analysis strategy. Values are mean values over 1000 simulated data sets. Additive error structure

dependencies within the field trial irrespective of the magnitude of the spatial variation (Rodríguez-Álvarez et al. 2018; Selle et al. 2019). It is worth mentioning that, for the linear spatial covariance models considered, including linear covariance and P-splines, spatial models across and within replicates will yield the same results (Williams et al. 2006; Boer et al. 2020).

Effectiveness of spatial models across designs and trends

Like previous work, it was found that models with spatial trend adjustment resulted in lower bias and higher correlations across designs, but that the improvement was smaller than what could be achieved by an appropriate choice of experimental design. Borges et al. (2019) found that spatial correlation models with a one-dimensional autoregressive process or a two-dimensional exponential process improved the performance of a completely randomized design, a RCBD, an alpha lattice design and a partially replicated design, but that the use of a spatial model did not outperform the experimental design. In the present work, it was expected that the higher flexibility of the SpATS model (Rodríguez-Álvarez et al. 2018) would lead to an increased precision compared to the two other models with spatial correction. However, across field trends, this was mainly the case for the alpha and augmented designs and mainly in terms of correlation.

In the analysis of the three breeding trials, the SpATS model with row-column adjustment stood out by resulting in yield estimates deviating the most from the remaining models. This may lead to the suspicion that this model was not able to appropriately capture the variation in data. However, in the simulation study, the SpATS model with row-column adjustment had the lowest bias and highest correlation with the true yield for one of the two field trends from the breeding trials. Surprisingly, this model also performed best for the RCBD and augmented RCBD designs, which typically require a random effect structure based on the nested design (Jensen et al. 2018).

Absence of competition effects were assumed in the simulations. For small plots, this may not be a realistic scenario. In large breeding trials, generally the entries are related by pedigree. This makes a nested treatment structure, where entries can be grouped into families such that entries within groups are more related than between groups, interesting to the breeder due to the practical merit by easy comparison of entries with similar phenotypes. There are three ways to place families of entries in the field. Relatives could be placed close to each other in the field, randomized in a way that separates families as much as possible or placed based on a full unrestricted randomization. In the present simulation study, an unrestricted randomization was assumed as recommended by Piepho and Williams (2006) who based on simulations showed that a split-plot type design where families were kept close together in the field is rarely an optimal strategy when selection is to be made across families. In addition, the simulation study assumed independence between entries. However, incorporating molecular markers into the models to capture relationships among entries is expected to enhance the precision of the models. By borrowing information from related entries, the use of molecular markers would effectively serve as additional replications, thereby increasing the accuracy of the results (Piepho et al. 2008; Heslot et al. 2015; Merrick et al. 2022).

The simulation study was carried out with two different effects of the field trend of the estimated trait representing two levels of heritability. Trait heritability did not change the overall conclusions from the simulation study. The relative performance of designs and analysis approaches was similar between heritability levels, while the precision as expected was higher for the low heritability. Similar findings were reported by Borges et al. (2019). It was, however, found that precision was more severely affected for one of the field trends (trend 5) in one combination of design and analysis strategy, emphasizing the benefit of incorporation previous knowledge of the field trend into the decision of design and analysis strategy.

Relative performance of designs

For field layouts using a rectangular grid of plots, row-column designs have been shown to be more efficient compared to designs with one-dimensional blocking, such as resolvable incomplete block designs like alpha-designs (Piepho & Williams 2006). This was supported by the present study, which identified the row-column design as the overall best-performing design. When analyzing individual field trends, only six of the field trends were best handled using the row-column design. For the two field trends with no or a systematic trend (5 and 6), one-dimensional blocking designs resulted in lower bias and higher correlation. Another simulation study found alpha designs to outperform row-column designs in terms of correlation to the true yield (Hoefler et al. 2020). This partially aligns with the findings from the present study, as their underlying trends did not involve as much variation as covered by some of the field trends considered here.

Improved efficiency of alpha lattice designs over RCBD in terms of lower coefficient of variation, mean squared error, and standard error of difference has previously been reported for maize (Khan et al. 2015), rice (Kashif et al. 2011), potato (Masood et al. 2008; Khan et al. 2015), and wheat (Masood et al. 2008; Khan et al. 2015; Borges et al. 2019). Overall, the present simulation study found that the settings with alpha designs resulted in lower bias, but also lower correlation compared to the RCBD designs. Furthermore, considering the field trends one by one, the best combination of design and analysis strategy in terms of correlation included the RCBD for five of the trends, including the two trends, 7 and 8, based on the breeding trials.

Among the designs considered, the two augmented designs generally resulted in the highest bias and lowest correlations to the true yield. This is in accordance with previous findings of fully replicated designs outperforming partially and un-replicated designs (Hoefler et al. 2020). Besides a higher uncertainty on the estimated traits, a general disadvantage of the augmented designs is that a big part of the plots is allocated to entries that are not of primary interest. In practice, however, breeders may prefer to replicate checks more often than entries to facilitate visual assessment of traits. This will result in a larger number of plots required, but at the same time, the standard errors of a difference among entries will be improved (Piepho et al. 2006). While the augmented designs did not prove to be the most efficient in the present settings, they have been shown to outperform replicated designs in multi-environment studies by providing a better description of the genotype x environment interaction (Moehring et al. 2014). An additional advantage of the augmented designs is that they provide larger population sizes for genomic mapping and selection. However, simulations have shown increased prediction accuracy in genomic selection for completely replicated designs over non-replicated designs (Hoefler et al. 2020).

In the simulation study, augmented designs based on RCBD and row-column designs were used, with the augmented row-column design demonstrating slightly better performance among the two. A systematic arrangement of check lines is an alternative option, but it has been shown that the actual arrangement of the check lines in the field is of less importance for the precision than the number of check lines (Clarke and Stefanova 2011). However, in this study, increasing the number of check lines from 3 to 5 only had a minor effect. Generally, it would be expected that it is the total amount of plots allocated to repeated entries that will have the biggest effect on the design performance through the influence on the total degrees of freedom (Clarke and Stefanova 2011).

Integrating UAV-derived field trends for informed trial planning

The present work builds on the idea that the general field trend is known before selecting the experimental design and analysis strategy (Cooper et al. 2014). Several statistical models include an adjustment for the spatial trend, but modeling the field trend is rarely a goal in itself. From the analysis of the three breeding trials, it was found that the field trends extracted from UAV campaigns during a growth season were correlated to the field trend in yield. Field trends could accordingly be estimated using a simple KNN smoothing of the residuals or as part of the SpATS analysis of vegetation index data.

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Author contributions Study conception and design was made by Signe M. Jensen. Breeding trial design and data collection was made by Rasmus Hjortshøj and Janus Asbjørn Schatz-Jakobsen. Data analysis were performed by David Redek and Jens Riis Baalkilde. The first draft of the manuscript was written by David Redek and Signe M. Jensen and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The code and datasets generated during the current simulation study are available from the corresponding author on request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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